Why Did American Student Achievement Decline?

Verifying the post-2013 fall and testing its explanations

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Motivation: the largest sustained decline on record

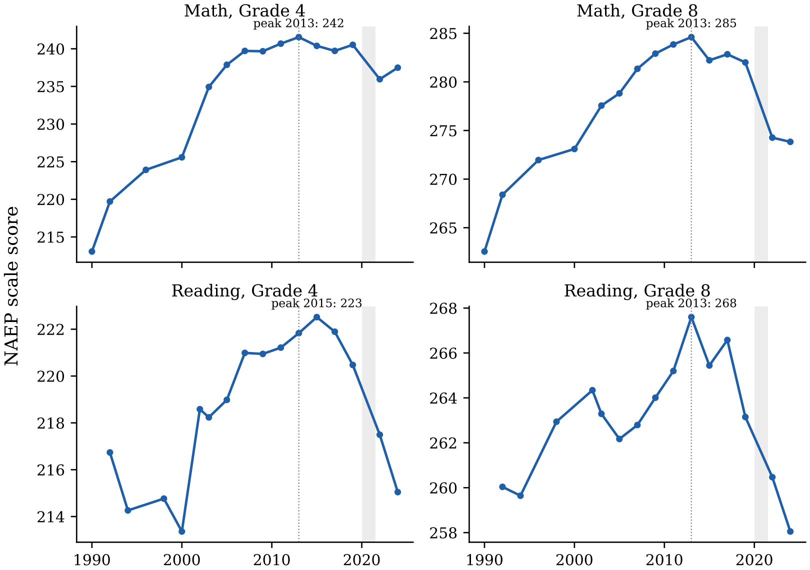
- U.S. achievement peaked around **2013** and has fallen since — seven years *before* COVID.
- By 2024: grade 8 math back to its **2000** level; grade 8 and grade 12 reading the **lowest ever recorded**; cumulative decline $\approx$ 0.14–0.30 SD ($\sim$ a full grade level at G8).
- 90–10 achievement gaps are the **widest in NAEP's history**.
- This talk: verify the facts from primary data, then **test every major explanation** against discriminating evidence — timing, distribution, cohorts, sectors, policy variation, international and adult parallels, and two causal designs.

All NAEP estimates pulled directly from the NCES Data Service API; replication repo with full audit trail.

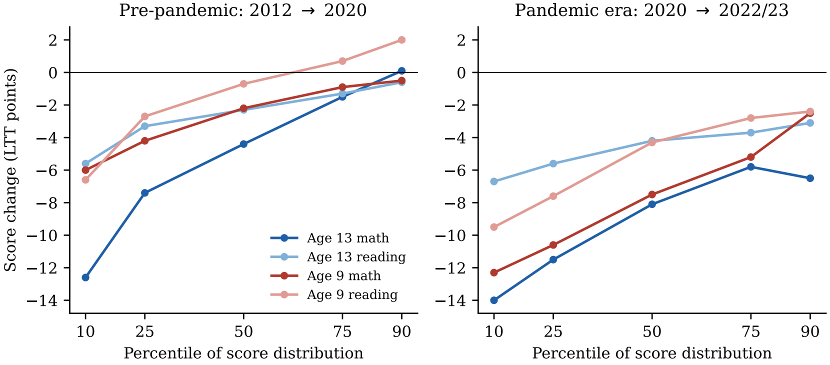
Preview of findings

1. The decline has **two phases with different causes**:
   - **2013–2019:** erosion confined to the *bottom* of the distribution;
   - **2019–present:** an unrecovered pandemic shock sustained by chronic absenteeism.
2. Direct tests reject or marginalize the standard school-policy suspects: demographics, funding, Common Core, NCLB-waiver deregulation, test effort, measurement artifacts.
3. The surviving phase-one explanation is **digital media saturation**: it alone fits the timing, the international and *adult* synchrony, the cohort structure, the sector neutrality, and the microdata exposure profile.
4. Two original causal exercises: a **waiver event study** (precise null) and the **first within-U.S. 4G-rollout design on test scores** (precise null on arrival timing — informative about mechanism).

Fact 1: every national series peaks ~2013, two distinct drops



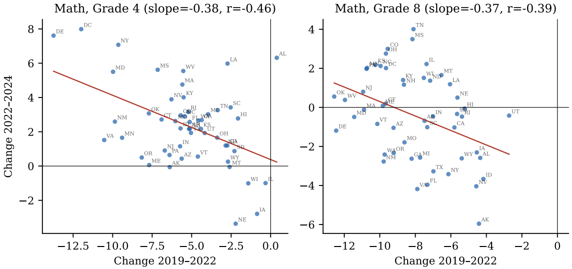
Fact 2: phase one is a collapse *at the bottom*



- Age-13 math, 2012→2020 (pre-pandemic): P10 −12.6 vs. P90 +0.1.
- 2022–24: the top recovers; the bottom keeps falling. G8 math 90–10 gap: 93 → 109.

NAEP Long-Term Trend; main-NAEP percentile series confirm (PC:P1–P9).

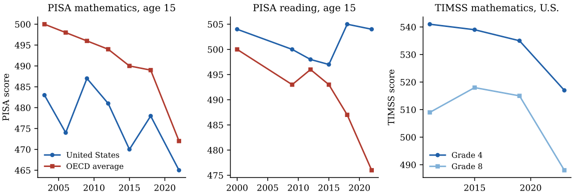
Fact 3: declines occur *within* every group, state, and sector



- Falling within every racial/ethnic and income group — composition explains ≤1–2 points.
- Every state below its 2013 level in G8 math.
- Same trend in Catholic schools (more below).

NAEP main assessment, state panels and subgroup series.

Fact 4: not just American, and not just children



- PISA: synchronized pre-pandemic declines across rich countries with very different school policies.
- **Adults** (PIAAC): U.S. literacy −12 2017→2023; share at/below Level 1: 19%→28%; top stable. Declines in 19 OECD countries.
- Same bottom-heavy signature everywhere.

OECD PISA 2022 Vol. I; PIAAC Cycle 2 (caveats: 28% U.S. response rate, tablet-only).

The facts any explanation must fit

1. Onset ∼2013, well before COVID — timing
2. Concentrated at the bottom decile; top flat-to-rising pre-2020 — distribution
3. Within every demographic group and state — not composition
4. Uniform 2019–22 drop; post-2022 recovery only at the top — two phases
5. Present in Catholic schools — not school policy?
6. Synchronized internationally — not U.S. policy
7. Extends to adults — not schooling at all?
8. Accompanied by collapsing voluntary reading: 27% (2012) → 14% (2023)
   mechanism clue

Eight candidate explanations

Schooling-side

- H1 Pandemic disruption
- H3 Retreat from NCLB accountability
- H4 Great-Recession funding cuts
- H8 Teacher quality / grade inflation

Student/society-side

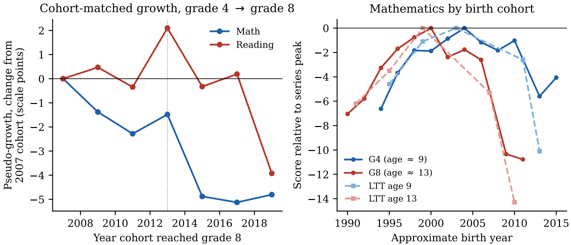
- H2 Smartphones & digital media
- H5 Demographic change
- H6 Chronic absenteeism
- H7 Declining reading practice

Strategy: derive *distinct predictions* from each across timing, distribution, cohorts, sectors, geography, international scope — then score all hypotheses against all facts.

Quick kills: demographics, funding, artifacts, effort

- **Demographics (H5):** declines occur within every group; compositional shift explains $\leq$1–2 points of a 10+ point decline. Rejected.
- **Funding (H4):** spending recovered after 2013 while scores kept falling; credible $-effects are an order of magnitude too small. Rejected as a primary cause.
- **Measurement artifacts:** exclusion flat at 1–3% (2013–2024); participation identical in 2022 and 2024. Ruled out.
- **Test effort:** PISA effort thermometer fell only 0.13–0.16/10 (2018$\rightarrow$2022) $\Rightarrow$ <5% of the decline; same decline on consequential state tests and among adults. Bounded.

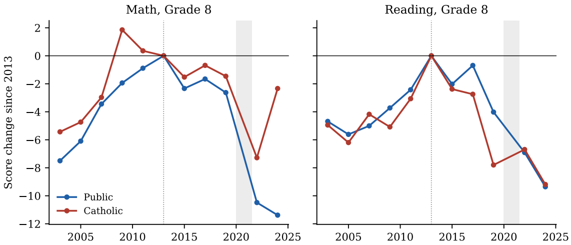
Test 1 — Cohorts: the losses accrue *during adolescence*



- Decompose every G8 change into **entry** (the cohort's G4 level) + **growth** (G4→G8).
- G8 reading 2013–19: $-4.4 =$ **entry** $+1.6$ − **growth** 6.0.
- Declining cohorts arrived at G4 at *record* levels, then lost ground in middle school: a period effect in adolescence, not a damaged-cohort story.

First formal cohort/period decomposition of the decline. Blind re-derivation reproduces it to the decimal.

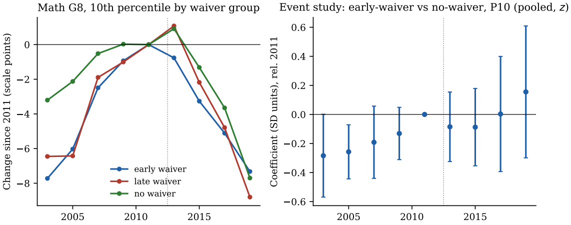
Test 2 — Sectors: Catholic schools declined in step



- Catholic schools were **never governed** by NCLB accountability.
- G8 reading 2013–19: Catholic −7.8 ($z = -3.4$) vs. public −4.0 — at least as large.
- Whatever drove phase one operated *across* the public/private boundary.

NAEP SCHTYPE series; published SEs. All-private reporting ends 2013, so Catholic is the usable benchmark.

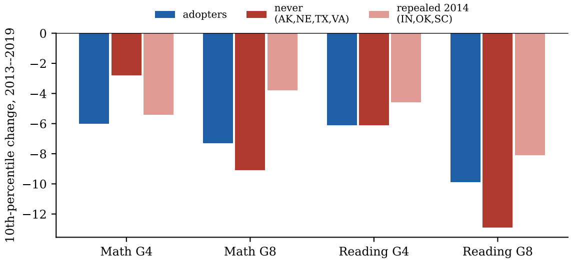
Test 3 — Policy variation: NCLB waivers (staggered event study)



- States exited NCLB accountability at **different times** (2012–15); seven never did.
- Dewey et al. (2026): this counterfactual has "never been tested."
- Pooled DiD on state P10: ≈0, **permutation** $p = 0.46$.
- Informative null: $\mathbf{MDE_{80}} \approx \mathbf{3.2}$ **points** — under half the Dee–Jacob NCLB effect.
- Caveat: joint pre-trend $p = 0.08$.

Randomization inference, 2,000 permutations. Bleiberg (2020) binary/means version through 2013: also null.

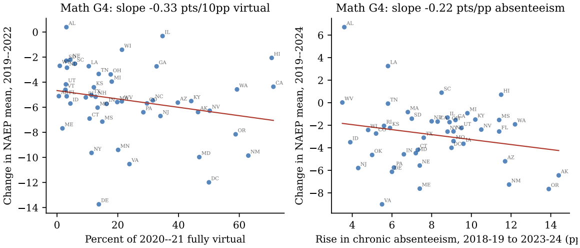
Test 4 — Common Core: never-adopters declined anyway



- AK, NE, TX, VA never adopted; their P10 declines are $\geq$ adopters' (diff -0.4, $p = 0.72$).
- Minnesota (math-only non-adoption): its *non-CC* subject fell *less* — wrong sign for CC harm.

State adoption/repeal coding; triple-difference for MN.

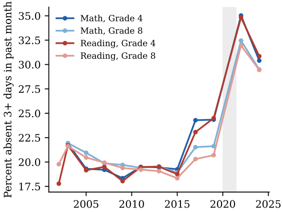
Phase two: the pandemic shock is real — but aggregation matters

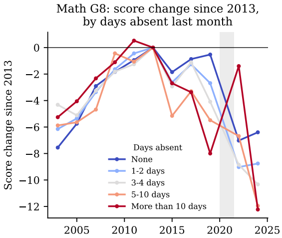


- District-level closure effects are decisive (Goldhaber et al.; Jack–Oster).
- But **between states**, 2020-21 virtual share explains ≈nothing of NAEP changes ($-0.04/10\text{pp}$, $p = 0.80$): state aggregation washes out district treatment variance.
- The puzzle is not the drop — it's the **non-recovery**.

CSDH schooling-mode shares × state NAEP changes; TUDA 26-district version similar (wide CIs).

The recovery brake: absenteeism — with a pre-pandemic warning

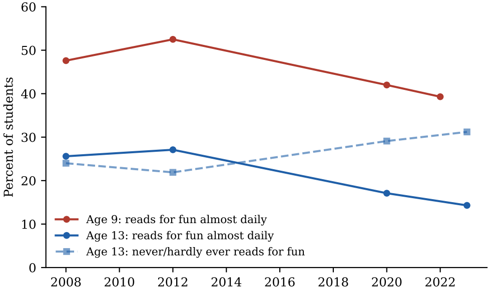




- NAEP's own absence item: 3+ days missed last month rising *before* COVID (G4: 19.5% → 24% by 2019), then 32–35% in 2022, ∼30% in 2024.
- Kitagawa: the absence shift accounts for **25–42% of 2019–24 declines** — but only **7–19%** of pre-pandemic G8 declines.
- Phase one was not an attendance phenomenon; phase two partly is.

B018101 crosstabs; decomposition is compositional, not causal. Blind re-derivation matches exactly.

The surviving hypothesis: digital media (H2) — timing and mechanism



- Teen smartphone ownership: 23% (2011) → 73% (2015) → 95% (2018). The peak-and-decline sits exactly on the takeoff.
- Reading for fun (age 13, almost daily): 27% → 14%, *mostly pre-pandemic*.
- H2 is the only hypothesis that natively explains the international *and adult* synchrony.

Pew; NAEP LTT student survey (S003501).

H2 in microdata: exposure sits exactly where the damage is

PISA 2022, 613,744 students; Rubin-combined PVs, school-clustered, ESCS-adjusted:

- Classroom device distraction (most/every lesson): −13 (U.S.) / −15 (OECD) points — survives SES adjustment essentially intact. 5+ hrs leisure device use: −40.
- Within countries, every over-engagement measure is **monotonically concentrated in the bottom score quartile** (distraction 32% vs. 24%; device anxiety 24% vs. 12%) ...
- ...yet **nearly flat across SES quartiles** (27.9% vs. 25.8%): a low-*performer* phenomenon, not a low-*income* one — matching the decline's signature in a way no compositional story does.
- Associational, two-way causality plausible; what it establishes is that the exposure profile and the damage profile coincide.

ST273Q06JA coding verified from file metadata (and independently re-derived blind). Item is rotated: U.S. $n \approx 3{,}000$.

H2 causal corroboration from adjacent literatures

School phone bans (quasi-experimental):

- England: $+0.064$ SD, and $+0.142$ SD for low achievers, ≈ 0 for high — the mirror image of the decline (Beland–Murphy). Norway, Spain: similar; Florida 2023: gains by year two; Sweden's null where rules already bound.

Fertility (new 2025–26 rollout designs, same shock):

- iPhone access (AT&T-exclusivity design) cut teen births 4.5–8% (Myers–Hooper); 4G arrival (ruggedness IV) collapsed teen fertility, raised teen suicides (Hudson–Moscoso Boedo).
- Effects **monotone-declining in age, null for 25+**: the critics' objection to the baby-bust claim is exactly our prediction for an adolescent learning shock.

Cross-country: 3G arrival $\times$ PISA, 82 countries: -0.04 to -0.08 SD (Jain–Stemper).

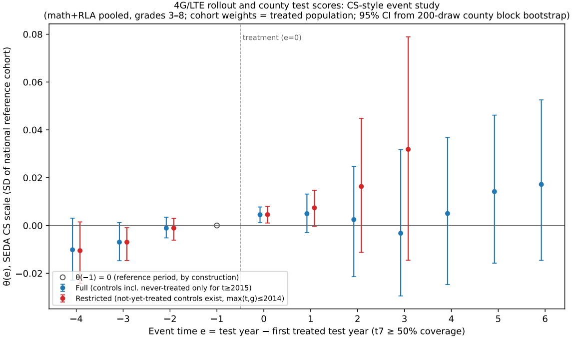
New test: the first within-U.S. mobile-rollout design on test scores

Design. Archived National Broadband Map: block×provider mobile coverage, 9 semiannual waves spanning the entire 2010–14 LTE rollout (validated against FCC benchmarks and Verizon's Dec-2010 launch markets). County treatment = first wave coverage >50% of population. Outcomes: SEDA 5.0 county means, grades 3–8, 2009–2019 (99.9% merge).

- Identifying variation = the **rural lag**: metro counties treated ∼2011–12, nonmetro 1–3 years later.
- Estimators: Callaway–Sant'Anna event study; continuous dose (county FE + state×year×grade×subject FE); within-county-year **adolescent-exposure gradient** across grades; permutation inference.

Wyckoff (2025): direct causal smartphone→achievement evidence "lacking." Free data end to end; full pipeline in repo.

4G result: a precise null on the arrival-timing margin



- Dose: -0.002 **SD** (perm $p = 0.66$; $\mathbf{MDE}_{80} = 0.013$). Gradient: $+0.011/\text{yr}$ (perm $p = 0.12$; MDE 0.016).
- 10-spec robustness grid: all $|\hat{\beta}| \leq 0.003$, signs flip. Pre-trend drift disclosed.
- Reading: the decline did not ride on local cell-tower timing. The national, adoption-driven channel is invisible here (year FE; 3G-era adoption; controls gone by 2015).
- Contrast: the *same* variation moves teen fertility — not test scores.

Synthesis: a dual-criterion test

Borrowed from the fertility-puzzle debate (Kearney–Levine; Evans): a candidate cause must clear **two bars at once**:

1. **Synchrony:** explain the same decline across countries with disparate school systems — and among adults.
2. **Signature:** explain bottom-of-distribution concentration, adolescent-period timing, sector neutrality.

- Every school-policy explanation fails bar 1 by construction.
- The pandemic fails bar 2 for phase one (it arrived six years late).
- **Digital media is the only candidate examined that clears both.** Accountability retreat retains a marginal supporting role (G4 public bottom decile); absenteeism owns the phase-two non-recovery.

Scorecard

	Timing	Bottom-heavy	COVID drop	No recovery	Within-grp	Int'l
H1 Pandemic	$\times$	$\sim$	$\checkmark$	$\sim$	$\checkmark$	$\sim$
H2 Digital media	$\checkmark$	$\sim$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$
H3 Accountability	$\checkmark$	$\checkmark$	$\times$	$\sim$	$\checkmark$	$\times$
H4 Funding cuts	$\sim$	$\times$	$\times$	$\times$	$\sim$	$\times$
H5 Demographics	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$
H6 Absenteeism	$\times$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\sim$
H7 Reading practice	$\checkmark$	$\sim$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$
H8 Teachers/inflation	$\sim$	$\sim$	$\times$	$\sim$	$\sim$	$\times$

Two-phase accounting: phase one $\approx$ 20–45% of the total 2013–24 decline; pandemic window 30–70%; post-2022 the rest. Full COVID recovery would still leave the U.S. far below 2013.

What would settle it: a pre-registered test arriving in 2027

- $\sim$30 states adopted school phone bans between 2023–24 and 2025–26, with staggered timing and varying bindingness (bell-to-bell vs. instructional-time).
- **Pre-registered design** (frozen before data exist): exposure groups from 50-state statute coding; NAEP 2026 state percentiles; same permutation machinery as the waiver test.
- Predictions on record: gains at P10/P25, larger under bell-to-bell bans; top decile $\approx$ flat.
- This is the waiver test's logic pointed at the leading hypothesis — with $\sim 4\times$ the treated states.

Also on the docket: restricted-use NAEP microdata; PISA 2012→2022 ICT panel; SEDA/adoption-margin designs.

Conclusion

1. The decline is real, large, verifiable — and **older than COVID**.
2. It is a crisis of the **struggling student**: bottom-decile collapse, record gaps.
3. Direct tests reject the school-policy suspects; the evidence triangulates to **digital media** for phase one and **absenteeism-sustained pandemic damage** for phase two.
4. Policy implications: treat attendance as the front line; take *binding* phone restrictions seriously (2026 will tell); target the bottom of the distribution; and do not mistake "back to 2019" for recovery — 2019 was already pointing down.

Paper, data, code, claims audit, and pre-registration:
github.com/brendanbartanen-svg/naep-achievement-decline

Appendix: verification layer

- **Claims audit:** 41 load-bearing claims, each typed (citation/pulled/computed) with a $<$5-minute verification path; 62-assertion test suite runs clean.
- **Blind clean-room replications** (independent agents, no access to analysis code, fresh API/microdata pulls):
  - Cohort decomposition: exact match ($-4.45 = +1.57 - 6.02$).
  - PISA distraction: exact U.S. match (-13.25); item coding independently confirmed from file metadata.
  - Kitagawa absence: data match to the decimal.
  - Waiver: null + MDE confirmed; point estimate spec-dependent (≈ 0).
- **External anchors:** waiver null $\leftrightarrow$ Bleiberg (2020); state-aggregation null $\leftrightarrow$ Goldhaber/Jack–Oster; sector pattern $\leftrightarrow$ Malkus/Petrilli; PISA gaps $\leftrightarrow$ OECD published bivariates.

Appendix: 4G robustness grid

Specification	Coef. (SD)	p (cluster)
Dose, t7 (primary)	−0.0022	.57
Binary first_t7_50	−0.0017	.58
Binary first_t6_50	−0.0030	.35
Binary first_t7_25	−0.0022	.52
Binary first_t7_75	−0.0029	.28
Binary t6_50, Verizon-only	−0.0010	.81
Dose, drop artifact states	+0.0015	.64
Binary t7_50, drop artifact states	+0.0013	.63
Dose, unweighted	+0.0004	.92
Binary t7_50, unweighted	+0.0007	.81

County FE + state×year×grade×subject FE; SEs clustered by state; permutation inference within state.

Appendix: avenues examined and set aside

- **Opioid exposure:** state correlations null/inconsistent — tested and cut.
- **Cannabis legalization, discipline reform:** no clean variation — documented set-asides.
- **Cross-country smartphone-adoption timing:** data quality insufficient.
- **PISA leisure-hours items (ST326/ST322):** not administered in the U.S.; the IC171 fallback is an access artifact in a 95%-saturated population — cut.
- **TUDA district dose-response:** weak gradient, wide CIs — limitation, not contradiction, of district-level closure findings.